

LINGYU XU

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Education

Worcester Polytechnic Institute, Worcester, MA

Jan 2024 - Dec 2025

Master of Science in Information Technology in System Design & Artificial Intelligence

- GPA: 3.81
- **Coursework:** LLM, Application Development, Machine Learning, Data Mining, Data Visualization, SQL

Pennsylvania State University, State College, PA

Aug 2015 - May 2020

Bachelor of Science in Computer Science

- **Coursework:** System Programming, Network Programming, Algorithm, Data Structure, Statistics

Experience

Research Assistant

Oct 2025 - Current

Worcester Polytechnic Institute

- **CV-Arena** [🔗](#): Developed the Pro-Edit dataset and Resolution Baseline Agent to normalize evaluations. Introduced a secure and scalable benchmark and a feedback-loop agentic architecture strategically.
- **TimePre** [🔗](#): Developed a probabilistic forecasting framework with multi-hypothesis density estimation and a channel-based reversible normalization method to improve model stability, mitigate distribution shifts, and preserve long-horizon non-stationary patterns during de-normalization.

Software Development Engineer

Mar 2021 - Dec 2023

Dell Technologies

- Built automated **Python/Go/Java** APIs for **Azure HCI** Day-1 provisioning, covering Windows domain joining, security hardening, cluster validation, and VMware/VxRail pre-validation, reducing manual configuration and accelerating deployment timelines by 20%.
- Developed a user-friendly **React/Angular/TypeScript** interface for Day-1 operations, using dynamic forms, validation flows, and clear data visualizations to guide operators through server pre-configuration and reduce setup errors by 15%.
- Led a 4-person team to build a **Jenkins**-based CI/CD pipeline and containerized provisioning services with **Docker/Kubernetes**; integrated **MongoDB** and **PostgreSQL** APIs for configuration storage, validation, and retrieval, reducing service startup failures by 25% and manual testing effort by 30%.

Software Development Engineer Internship

Mar 2021 - May 2021

Dell Technologies

- Engineered **Python/Java** APIs to automate the Day-1 setup for Dell VxRail HCI systems, focusing on VMware configuration pre-validation, VxM network setup, and **PostgreSQL** table management. This automation initiative reduced incompatible configuration errors during deployment by 15%.
- Engineered responsive web interfaces with **Angular** for the VxRail Day 1 & Day 2 operations team. Developed UI components (dynamic forms, data filters) to pre-config server configuration for VLAN, authentication and backup settings, reducing manual setup steps and increasing team configuration efficiency by 10%.

Projects

Agentic Advising Assistant, Worcester, MA

- Multi-Agent System Architecture: Architected a production-grade Agentic AI advising platform using **FastAPI**, **LangChain**, and **Google Gemini 2.5 Pro**, orchestrating an Intent Classification Agent and a **ReAct** Adviser Agent over a **Hybrid RAG** pipeline (**SQLite** + **ChromaDB**) to automate complex academic advising workflows.
- Core LLM & RAG Engineering: Built a multimodal PDF extraction pipeline via **Gemini Vision API** for unstructured transcripts, and designed a DB-first hybrid retrieval layer with course-aware chunking, score boosting, and query-rewrite fallback across **SQLite** and **ChromaDB** vector store, eliminating hallucinations on course/program lookups.
- Performance & Productionization: Optimized the ReAct reasoning loop with 7 specialized agent tools, real-time **SSE** streaming, and **Redis** session memory (TTL + LRU eviction); containerized with **Docker Compose** (frontend + backend + Redis), achieving ~10s avg response latency and a projected 60% reduction in manual advising workload.

Android App Development, Worcester Polytechnic Institute, Worcester, MA

- Developed an **Android** application in **Java** and **JavaScript** simulating core banking operations, including balance inquiry, transaction history filtering, and dynamic sorting, backed by a structured **SQLite** database for persistent local storage.
- Engineered a normalized relational schema with interconnected tables; automated data generation using **Python** scripts and implemented backend data handling and query logic in Java to optimize retrieval performance and application responsiveness.

Skills

Languages: Python, Go, Java, JavaScript, TypeScript, C/C++, SQL

Web & Backend: React, Angular, Node.js, Express, FastAPI, REST APIs, GraphQL, HTML5, CSS

Databases & Cloud: PostgreSQL, MongoDB, MySQL, SQLite, Azure, Docker, Kubernetes, Jenkins, Git

AI/ML & Data: PyTorch, TensorFlow, Scikit-learn, Pandas, Spark, Tableau, Excel